An Effective and Lightweight Hybrid Network for Object Detection in Remote Sensing Images

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***Abstract*— In recent years, general object detection on nature images leverages convolutional neural networks (CNNs) and vision transformers (ViT) to achieve great progress and high accuracy. However, unlike nature scenarios, remote sensing systems usually deploy a large number of edge devices. Therefore, this condition encourages detectors to have lower parameters than high-complexity neural networks. To this end, we propose an effective and lightweight detection framework of hybrid network, which enhances representation learning to balance efficiency and accuracy of model. Specifically, to compensate low precision caused by lightweight neural networks, we design a boundary-aware context (BAC) module and a frequency self-attention refinement (FSAR) module to improve detector performance in a hybrid structure. The BAC module enhances the local features of the object by fusing the spatial context information with the original image, which not only improves the model accuracy but also effectively solves the problem of multiscale objects. To alleviate the interference of complex background, the FSAR module adopts an adaptive technology to filter out redundant information at different frequencies to improve overall detection performance. The comprehensive experiments on remote sensing datasets, i.e., NWPU VHR-10, LEVIR, and RSOD, indicate that the proposed method achieves state-of-the-art performance and balances between model size and accuracy.**

***Index Terms*— Efficient and lightweight, hybrid network, object detection, remote sensing images.**

I. Introduction

REMOTE sensing object detection is an essential task in the field of computer vision. It plays an important role in various applications, such as urban planning, land use, and disaster management. With the increasing availability of high-resolution remote sensing images, the need for lightweight and efficient object detection models becomes more and more urgent.

In the past decade, deep-learning-based object detection models have mainly fallen into two major categories: CNN-based object detectors and transformer-based object detectors. Mainstream object detectors based on convolutional neural networks (CNNs) are typically divided into two major categories: anchor-based object detection models, such as two-stage detectors region convolution neural networks (RCNNs) and its variants [1], [2], [3], [4], [5] and single-stage detectors YOLO series [6], [7], [8], [9], [10], [11]. These models typically use predefined anchor boxes to locate and identify objects in images. The second category includes anchor-free object detection models, such as CenterNet [12] detector. In contrast to the traditional CNN-based object detectors, transformer-based detectors have gained attention in recent years due to their outstanding capability to represent global features, such as vision transformer (ViT) [13], Swin transformer [14], [15], and others. With the development of CNN and transformer technologies, researchers have been exploring how to fully harness their respective strengths, leading to the emergence of hybrid structural model frameworks. For example, Zhang and Yang [16] introduce an efficient multiscale transformer called Rest, which combines CNN and transformer by introducing global features into the transformer and constructing efficient memory multihead self-attention (SA) mechanisms. In addition, Yuan et al. [17] develop a visual transformer based on a hybrid structure known as T2T-VIT to address the limitations of transformers in terms of global dependencies. Furthermore, Peng et al. [18] propose a hybrid network structure based on the conformer, primarily leveraging convolutional operations and SA mechanisms to enhance the model's learning capabilities. However, despite these hybrid structural models overcoming many of the shortcomings of CNN and transformer, their complex structures and extensive computational requirements pose challenges for practical application on edge devices. At the same time, these models exhibit exceptional performance on natural images, indicating their potential in the field of remote sensing object detection.

There are two main challenges in migrating detectors from natural scenes to remote sensing images. The first challenge is the difference in image attributes between remote sensing scenes and natural scenes. Compared with nature images, remote sensing images contain more complex characteristics, including dense object distribution, high resolution, confused background, arbitrary orientation, scale variance, etc. This difference prompts the remote sensing detectors to pursue more excellent solutions. Another one is that remote sensing devices usually require deployable detection algorithms with fewer parameters and higher efficiency than natural scenarios. As we all know, remote sensing systems usually deploy a large amount of edge devices, so it is difficult to provide enough computing power to support large-scale detection algorithms.

Manuscript received 28 August 2023; revised 16 October 2023 and 24 November 2023; accepted 2 December 2023. Date of publication 5 December 2023; date of current version 13 December 2023. This work was supported in part by the National Natural Science Foundation of China under Grant 62372348 and Grant 61976166, in part by the Shaanxi Outstanding Youth Science Fund Project under Grant 2023-JC-JQ-53, and in part by the Fundamental Research Funds for the Central Universities under Grant QTZX23057. *(Corresponding author: Weichao Yang.)*

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Digital Object Identifier 10.1109/TGRS.2023.3339624

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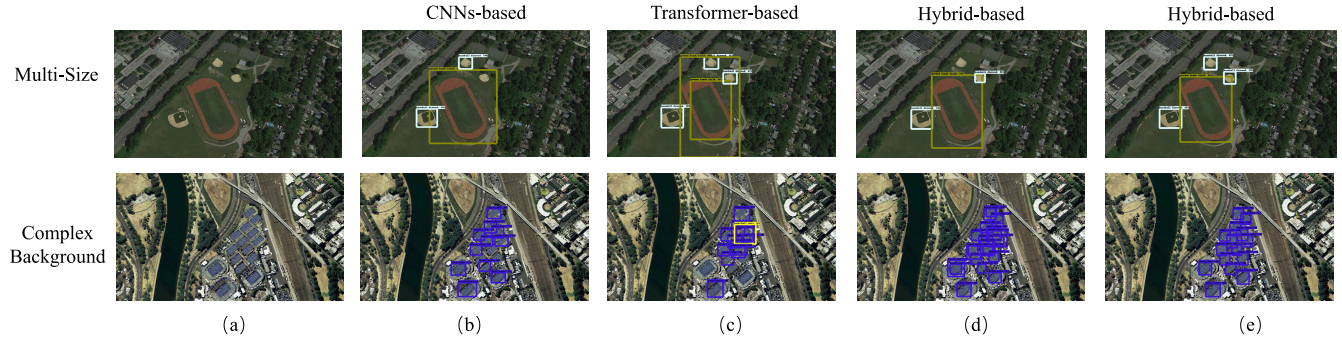


Fig. 1. Comparison of object detection results in remote sensing images using different lightweight models. In addition, prediction boxes in different colors correspond to different categories. i.e., light gray for baseball diamond, brown for ground track field, dark blue for tennis court, and yellow for basketball court. (a) Raw image. (b) MobileNetv2. (c) MobileVit. (d) Conformer. (e) Ours.

Therefore, researchers need to design lightweight detectors to meet the requirements of remote sensing devices. In brief, remote sensing detectors request to meet both efficiency and accuracy.

To solve the aforementioned limitation, researchers propose some lightweight models for remote sense object detection. Among the CNN-based approaches, SqueezeDet [19] is a lightweight single-stage detector that uses a fire module to reduce the number of parameters. Tiny-deeply supervised object detector (DSOD) [20] adopts depth-separable convolutional operators to effectively decrease computational complexity, achieving a lightweight remote sensing detector. Different from single-stage detectors, Pelee [21] is a two-stage detector that uses a novel structure to reduce the number of parameters. Recently, transformer models have been a success and attracted a lot of attention. As popular as transformer in natural scenarios, there are many variations in transformer structures in remote sensing scenarios. The deformable detection transformer (DETR) [22] introduces deformable attention to reduce the memory footprint while maintaining detection accuracy. Another lightweight transformer-based detector is SETR [23], which presents a spatially extended transformer to capture global contextual information and spatial dependencies.

Although various attempts have been made at lightweight remote sensing object detection, the existing methods perform inadequately in applications (as shown in Fig. 1). The CNN-based models show missed detections because the model only uses local features for object detection, resulting in insufficient target feature extraction in the challenging scenes with multiscale objects and complex backgrounds. Similarly, there also exist missed and false detections in transformer-based models. It is due to the influence of similar global features between object and environment as well as objects in complex backgrounds. For the existing hybrid architecture, the combination of local and global features reduces the missed and false detections [see Fig. 1(d)], but still fails to detect all the objects.

The lightweight networks based on CNNs, transformer, and the existing hybrid structures cannot effectively balance the efficiency and accuracy of the detectors. The main issue is that these lightweight networks are not able to availably perform object detection in case of dense multiscale objects, which leads to emergence of erroneous prediction boxes. However, the problem needs to be solved without introducing additional computational parameters. Inspired by this problem, this article proposes an effective and lightweight hybrid network for object detection in remote sensing images. On one hand, we present a boundary-aware context (BAC) module reprocessing the original image via the Sobel operator to generate the corresponding texture features, which are integrated with the original image to enhance the edge cues of the model. By embedding the Sobel operator, the BAC module elevates the perceptive ability of the objects and solves the aforementioned multiscale variances. On the other hand, the frequency SA refinement (FSAR) module is designed to use an adaptive technology for filtering out interference information at different frequencies, in which the different and confused background environments are adequately addressed. It can be observed that the proposed model can precisely and entirely detect each object in a given remote sensing image, indicating that our model has a superior feature extraction capability and can distinguish different features of different objects [as shown in Fig. 1(e)].

The proposed model not only enhances local and global feature representations through convolutional operations and SA mechanisms but also addresses the inherent challenges of remote sensing images. Unlike traditional model pruning, we start from scratch to design a lightweight neural network structure tailored to specific task requirements, avoiding complex pruning. We emphasize the overall model structure and parameter selection to achieve a balance between performance and complexity. In addition, we may incorporate lightweight strategies such as depthwise separable convolutions and channel attention mechanisms to reduce complexity while maintaining performance. In this article, extensive experiments on the NWPU-10 [24], LEVIR [25], RSOD [26], and DOTA [27] datasets validate the effectiveness and stability of the proposed model.

The main contributions of this article are outlined below in brief.

1) We propose an effective and lightweight detection framework of hybrid network for remote sensing images, which combines the advantages of convolutional operations and SA mechanisms.
2) We propose the BAC module to solve the multiscale problem of remote sensing images by extracting the target spaces' context information.

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3) For complex backgrounds, the FSAR module is introduced to refine the global features by filtering different frequency information.

The remainder of this article is organized as follows. In Section II, the remote sensing object detection and lightweight object detection are reviewed. Section III presents the proposed effective and lightweight hybrid network. The experimental results and analyses are presented in Section IV. Finally, the conclusion is drawn in Section V.

II. Related Work

A. Remote Sensing Object Detection

Over the past decade, there has been an increasing interest in object detection for remote sensing images. With the rapid development of deep learning, researchers focus more on CNN-based remote sensing object detection [28], [29], [30], [31].

To address the problem of multiscale objects in remote sensing images, decoupled classification localization network (DCL-Net) [32] enhances the holistic feature information by passing the information of low-level features from top to bottom. Unfortunately, this complicated design introduces noise and redundant cues. To alleviate this issue, multi-feature pyramid network (MFPNet) [33] integrates local and global features to fully fuse global and local information at all the scales for detecting discrete objects and objects with scale differences.

In addition, remote sensing images usually contain complex background, which causes serious disturbances in object detection. Shi et al. [34] introduce a spatial attention mechanism to adaptively incorporate contextual information into the feature maps to improve recognition and localization accuracy. To establish dependencies between channels, squeeze-and-excitation network (SENet) [35] uses a squeeze operation to capture relationships between features to emphasize useful features and suppress useless ones. Furthermore, RADet [36] models the spatial location correlation between global pixels and detects targets surrounded by confused backgrounds via joint spatial attention and channel attention.

In recent years, transformers have also been increasingly adopted in remote sensing object detection [37], [38], [39] as encoders to extract global contextual features. Adaptive dynamic transformer detection (ADT-Det) [40] enhances the semantic information of feature maps by embedding the transformer into a lightweight feature pyramid network (FPN) to connect features at different levels. To further improve the ability of extracting contextual information from model, adjacent context coordination network (ACCoNet) [41] proposes an adjacent context coordination module and a bifurcation aggregation module to capture the local and global features of the object.

However, the above models focus more on accuracy and neglect to balance model size and performance, hindering their deployment in resource-limited environments.

B. Lightweight Object Detection

Considering the resource constraints of edge devices, the existing complex and robust object detection models do not meet the actual resource requirements. Therefore, lightweight object detection is gradually gaining more attention. According to the type difference in neural network, they are divided into three main categories, i.e., lightweight CNN-based object detectors, lightweight transformer-based object detectors, and lightweight hybrid architectures' object detectors.

1) Lightweight CNN-Based Object Detectors: This kind of model replaces the complicated backbone network in the object detection model with a lightweight network [42], [43], [44], [45], [46] and fine-tunes the model structure.

GhostNet [47] proposes the Ghost module, which uses linear convolution with few parameters to extract similar features and performs feature reuse, greatly reducing the computations of the model. Li et al. [48] design the R-CNN as a lightweight network inspired by the inception module and the jump connection. To further reduce parameters, Zhou and Piao [49] use depth-separable convolution instead of standard convolution to construct a feature extraction network, which effectively reduces parameters while ensuring model accuracy but still does not meet the practical needs. GiraffeDet [50] constructs a new lightweight feature extraction network and a flexible multiscale feature fusion structure, which obtains higher accuracy and efficiency.

Despite many attempts, there are still two problems. First, the design is not specific to remote sensing datasets. Second, the lightweight CNN-based detectors only focus on local features, which limits the performance of models.

2) Lightweight Transformer-Based Object Detectors: The perspective of transformer-based detectors is to leverage spare attention [51], [52] to decrease the computations and complexity of models.

Sparse transformer [51] adopts a downsampling way based on a fixed pattern that selectively calculates self-attentiveness instead of all the positions. Longformer [52] dynamically calculates attention based on the distance between positions when calculating SA. Iterative multi-scale feature aggregation (IMFA) [53] is the first general paradigm for efficient multiscale features in transformer-based object detectors. For efficiency, IMFA samples multiscale features from only a few key points within a few promising regions. Although there are many well-known works on lightweight remote sensing detectors, two issues still remain. First, the objects in remote sensing datasets are relatively dense and easy to miss using only transformer structures. Second, the lightweight transformer-based detector only focuses on global features and is easy to miss small targets.

3) Lightweight Hybrid-Based Object Detectors: Different from previous works, to complement the lack of global feature extraction capability for CNNs, several hybrid structures [54], [55] of CNN and transformer are purposed to achieve object detection on remote sensing images.

Conformer [56] is the first concurrent network structure to fuse features in an interactive manner. The structure not only naturally inherits the structural advantages of CNNs and transformer but also maximizes the ability to retain local features and global representation. Compared with conventional detectors, the conformer has lighter parameters. In addition, MobileVit [57] serially combines CNN and SA modules

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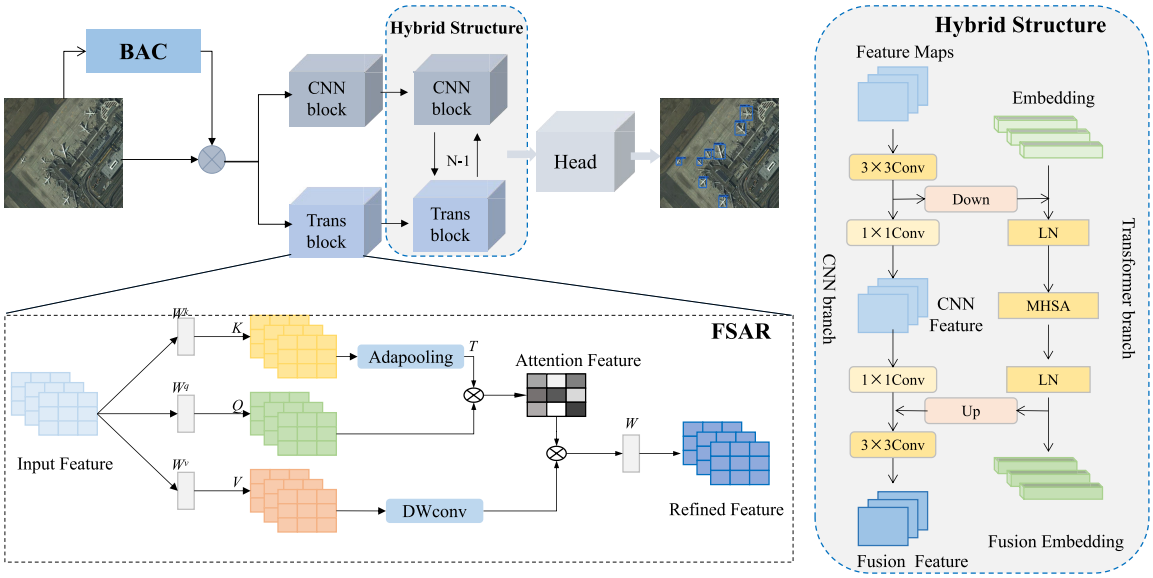


Fig. 2. Flowchart of the proposed model. First, for each input image, fusion features with contextual information extracted via the BAC module. Subsequently, the two-branch network extracts the local features and global representation of the object, respectively, where the FSAR module obtains useful low-frequency information through an adaptive pooling operation. Finally, the two parts of features, after feature alignment, are fed into the detection head for classification and regression. Besides, implementation details of the CNN block and Trans block are given. LN denotes layer norm and MHSA indicates multihead SA.

to extract features and obtain competitive results in object detection.

Unlike previous work, our approach to object detection in remote sensing images distinguishes itself by leveraging a combination of CNN and transformer-based hybrid architectures [54], [55], [56]. In remote sensing image detection tasks, we routinely face the challenges of complex backgrounds and multiscale objects. But CNN focuses more on global features, which is not ideal for complex backgrounds, while transformer focuses on global language information, which is not ideal for small targets. To address these challenges, our research endeavors to fully exploit the complementarity of CNN and transformer models for efficient information extraction and feature learning. Specifically, we combine CNN and transformer architectures to better capture spatial information in complex backgrounds and contextual information of multiscale objects. This combination is designed to achieve enhanced performance in remote sensing image detection tasks. By integrating CNN and transformer models, we aim to achieve a more comprehensive feature learning approach that caters to the specific requirements of various application domains.

III. Method

A. Overview of the Model

In the field of computer vision, local features and global features are widely used. In the era of deep learning, the traditional CNNs typically use convolutional operations to extract local feature mappings for object recognition. At the same time, transformers aggregate global representations using SA modules, embedding image patches into feature mappings.

To improve the representation of objects in remote sensing images, we improve the hybrid structure of the conformer and propose an efficient and lightweight remote sensing image fusion network, as shown in Fig. 2. Considering the multiscale and complex background characteristics of remote sensing images, we have designed two key modules: the parameterless border-aware context (BAC) module and the FSAR module. We use a lightweight dual-branch network, as our feature extraction network. In this network, we use depthwise separable convolution as the fundamental building block to construct the CNN branch and build the transformer branch using an SA mechanism. We achieve lightweight effects by reducing the number of CNN and transformer branches.

1) CNN Branch: To minimize the model's size, we use depthwise separable convolutions as the fundamental building block for constructing the CNN branch. The CNN branch uses a pyramid structure, where the network depth increases, the feature map's resolution gradually decreases, while the number of feature channels increases. The forward propagation process of the ith module in the CNN branch can be expressed as follows:

$$\widetilde{X_{c1}^{l}} = \mathrm{Conv}_{3\times 3}\left(\mathrm{Conv}_{1\times 1}\left(X_{c}^{l}\right)\right) \tag{1}$$

where X_c^l represents the input, $\widetilde{X_{c1}^{l}}$ represents the output, and $\mathrm{Conv}_{k\times k}$ denotes a convolution unit, where each convolution unit consists of a convolution layer with a kernel size of K.

2) Transformer Branch: We compress the feature maps generated by the stem module into 14×14 nonoverlapping patch tokens through a linear projection layer. Subsequently, we apply LayerNorm and a multihead SA module (MHSA). To ensure model lightweight, we use only two LayerNorm layers and one MHSA to construct the transformer branch. The specific mathematical representation is as follows:

$$X_{t}^{l+1} = \mathrm{LN}\left(\mathrm{MHSA}\left(\mathrm{LN}\left(\widehat{X_{t}^{l}}\right)\right) + \widehat{X_{t}^{l}}\right) \tag{2}$$

where X_t^l represents the input feature, X_t^{l+1} represents the output, LN stands for LayerNorm, and MHSA denotes the multihead SA.

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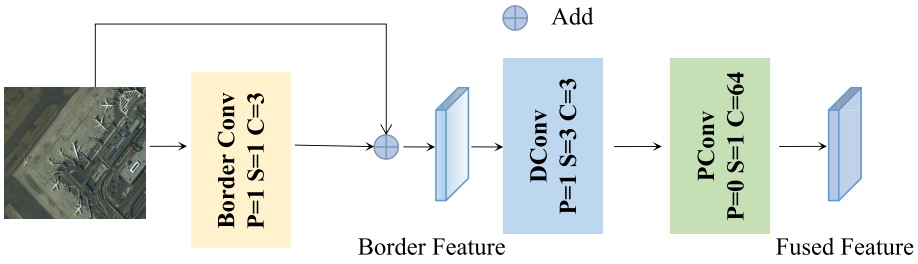


Fig. 3. Pipeline of BAC module. The border features are obtained by summing the feature maps after performing border convolution with the original image, and the fused features are obtained after depthwise convolution (DConv) and pointwise convolution (PConv) of the border features. In addition, P represents padding, S is the stride of convolution, and C indicates the number of channels.

The overall framework consists of the following steps. First, the BAC module extracts context-aware features from the input image and then fuses them with the original image features. Second, the fused feature mappings are fed into a two-branch network. The CNN branch extracts local features of the target using depthwise separable convolution, while the transformer branch extracts refined global feature maps through the FSAR module. Finally, the local feature maps extracted by CNN are channel-aligned and summed with the global feature maps from transformer using a 1×1 convolution, resulting in the final feature maps. These final feature maps are then input into the detection head for classification and regression.

B. BAC Module

Although the two-branch object detection network has the ability to extract global and local features of objects, the predicted coordinate regression in the detection task relies more on the contextual information of the image, such as boundary information, rather than the deep semantic features. This section proposes a BAC module, as shown in Fig. 3, which aims to improve the ability to extract spatial contextual information from images and enhance the localization capability of the model.

In contrast to other approaches, the extraction of border context information is first performed by boundary convolution with fixed parameters, considering the presence of multisize objects in remote sensing images. Here, the border convolution uses the Sobel operator extended to 3-D, which can be described as

$$B_{\text{conv}} = I * B_x + I * B_y + I * B_z \tag{3}$$

where I denotes the input image, B_x, B_y, and B_z denote the convolution kernels of the Sobel algorithm in the 3-D direction, $*$ denotes the convolution operation, and B_{conv} denotes the boundary convolution result.

Afterward, the border feature maps are added to the original image to obtain the enhanced border features, which are more conducive to recognition and detection by the feature extraction network, which can be formulated as

$$F_{\text{Border}} = I + B_{\text{conv}}. \tag{4}$$

Finally, a high-dimensional feature representation is achieved by updimensioning the convolution operation, so that the subsequent two-branch network can further extract different features, which can be formulated as

$$F_{\text{fused}} = D(F_{\text{high}})(P(F_{\text{high}}(F_{\text{Border}})) \tag{5}$$

where D and P denote the deep and pointwise convolution operations, respectively.

Considering the lightness of the model, depthwise convolution and pointwise convolution are used to reduce parameters of the model while learning the relationship between the upper and lower elements of the different channels. In this article, stride and channel of the depthwise convolution are set to 3 and 3, while the ones of the pointwise convolution are 1 and 64, respectively.

To demonstrate the effectiveness of the BAC module more vividly, we perform a visual comparison, as shown in Fig. 4. After embedding BAC module, we obtain richer contextual information, which is also commensurate with the fused image. The increase in texture information facilitates feature extraction at different scales, thus improving the effectiveness of the model.

C. FSAR Module

Unlike the well-established channel attention and spatial attention, the SA mechanism adjusts the weighting parameters only through the interaction of internal information. In view of the fact that object detection in remote sensing images is subject to complex background interference, the existing SA mechanism only performs the calculation of the similarity attention feature map through the self-similarity of the matrix and does not process the global information to achieve a more effective global feature extraction. As shown in Fig. 5(a), the SA module essentially calculates the similarity to the original feature map and gets a better feature representation by internal representation

$$\text{SA}(Q, K, V) = \text{softmax}\left(\frac{Q \cdot K^T}{\sqrt{dk}}\right) \cdot V \tag{6}$$

where Q, K, and V are the three matrices after the linear transform of embedding, T means transpose operation, softmax() represents the normalized score of the feature map, and $\sqrt{dk}$ denotes the dimension of the feature map, which is used to make the result of $Q * K^T$ satisfy the distribution with expectation 0 and variance 1, avoiding the saturation zone of the Softmax function.

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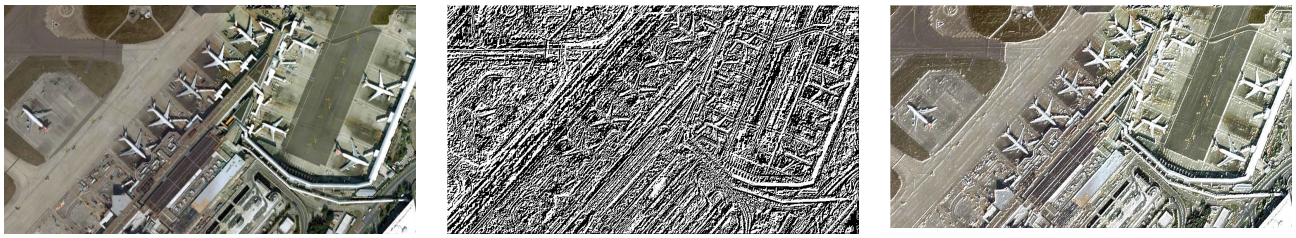


Fig. 4. Comparison of visualization results before and after implementing the BAC module.

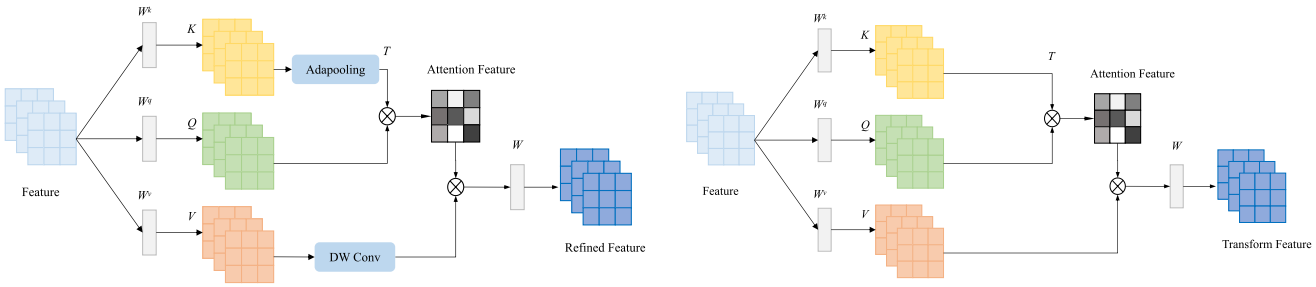


Fig. 5. Comparison of the traditional SA and the proposed FSAR. (a) SA. (b) FSAR.

As shown in Fig. 5(b), this section proposes an FSAR module, which introduces frequency-domain feature processing in the SA mechanism

$$\mathrm{FSAR}(Q, K, V) = \mathrm{softmax}\left(\frac{Q \cdot \mathrm{LK}^T}{\sqrt{dk}}\right) \cdot \mathrm{HV} \tag{7}$$

where LK represents low-frequency features, and HV denotes high-frequency features.

Concretely, channel-by-channel adaptive pooling is used as different low-frequency feature filters. The low-frequency matrix LK is extracted from the feature matrix K, i.e.,

$$\mathrm{LK}(p_c) = \sum_{1}^{c} \max_{i,i}(p) \tag{8}$$

where p represents the matrix of each channel, c indicates the number of channels, and i denotes the range of pooling steps. In the experiment, since the number of feature channels is 6, the number of channels is divided into two groups, and i is set as 1, 2, and 3, respectively.

In addition, the feature matrix HV is subjected to deep convolution to extract high-frequency features, i.e.,

$$\mathrm{HV} = \sum_{1}^{c} \mathrm{DWconv}(V) \tag{9}$$

where DWconv() denotes depthwise separable convolution.

D. Detector Loss

The overall loss of our framework consists of the classification loss, the regression loss, and the targeting loss. The classification loss can be expressed as

$$L_{\mathrm{cls}} = -\frac{1}{N}\sum_{i=1}^{N}\sum_{j=1}^{C} y_{ij}\log(\hat{y}_{ij}) \tag{10}$$

where N represents the number of samples, C is the number of categories, and y_{ij} is the true label (0 or 1) of the ith sample belonging to category j. The model's predicted probability of this sample belonging to category j is y_{ij}. $\hat{y}_{ij}$ is the model's predicted probability that the sample belongs to category j.

The regression loss can be expressed as

$$L_{\mathrm{box}} = \frac{1}{N}\sum_{i=1}^{N} \mathrm{SmoothL1}(b_i - b_{\mathrm{gt},i}) \tag{11}$$

where b_i is the position of the ith prediction frame predicted by the model, and $b_{\mathrm{gt},i}$ is the true target box.

The object localization loss can be expressed as

$$L_{\mathrm{loc}} = -\frac{1}{N}\sum_{i=1}^{N} \log\left(\frac{\mathrm{IoU}(y_i, b_{\mathrm{gt},i})}{p_{\mathrm{gt},i}}\right) \tag{12}$$

where p_i is the confidence of the ith prediction frame predicted by the model, and $p_{\mathrm{gt},i}$ is the confidence of the true target frame.

The overall loss can be expressed as

$$L_{\mathrm{total}} = \lambda_{\mathrm{cls}} \cdot L_{\mathrm{cls}} + \lambda_{\mathrm{box}} \cdot L_{\mathrm{box}} + \lambda_{\mathrm{loc}} \cdot L_{\mathrm{loc}} \tag{13}$$

where λ represents the hyperparameter that weighs the different loss terms.

IV. Experiments

A. Dataset

We verify and evaluate the effectiveness of our detector on three public remote sensing benchmarks, including the NWPU-10, RSOD, LEVIR, and DOTA datasets, which not only cover multiple complex scenes but also contain a variety of sizes and categories of objects. The NWPU-10 contains a total of ten classes of objects, namely, airplane, ships,

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TABLE I
EXPERIMENTAL RESULTS OF EACH COMPONENT ON THE NWPU-10 DATASET

Conformer	✓	✓	✓	✓
BAC		✓		✓
FSAR			✓	✓
$mAP(\%)$	91.04	93.12	92.56	94.12
$Params$(M)	24.24	24.24	24.24	24.24

TABLE II
COMPARISON RESULTS OF DIFFERENT CONVOLUTIONS

method	mAP**(%)**	$Params$**(M)**	$Flops$**(G)**
FFT	91.04	24.24	5.2
Canny	93.12	24.24	5.2
Robert	92.56	24.24	5.2
Sobel	94.12	24.24	5.2

storage tank, baseball diamond, tennis court, basketball court, ground track field, harbor, bridge, and vehicle. The RSOD dataset, an open remote sensing object detection dataset with 976 images and 6950 instances, contains four object types, i.e., aircraft, oil tank, playground, and flyover. The LEVIR dataset is made up of a large number of high-resolution 800 × 600 pixel Google Earth images and over 22k images, covering most of the ground features with complex and diverse backgrounds. For all three datasets, we use a random selection of 80% of the entire dataset as the training set and 20% of the entire dataset as the test set. The DOTA [27] dataset is a collection of data created. This dataset comprises 2806 large images with 188 282 instances covering 15 different object categories. The original images have a size of 4000 × 4000 pixels and are processed through cropping to 1024 × 1024 pixels with a 200-pixel overlap. Following the same setting from [58], we use half of the original images for the training set, one-sixth for the validation set, and one-third for the test set.

B. Implementation Details

We use the PyTorch framework to implement the proposed model on the Ubuntu V16.04 system. We apply Conformer-T as the benchmark feature extraction network. All the experiments are performed on one 2080Ti with 24 GB of memory. Pretrained conformer-tiny network weights are used to initialize the backbone network. During training, the input image is resized to 512 × 512. The training model is trained using the SGD algorithm with a momentum of 0.9 and 0.0005 weight decay. The initialized learning rate is set to 1e-4, the minimum value of the learning rate is set to 1e-6, and the learning rate is decayed using the cosine annealing algorithm. The whole training process has 200 epochs with a batch size of 8. Incidentally, to evaluate the performance of different models, the IoU value is constantly set to 0.5.

C. Ablations Study

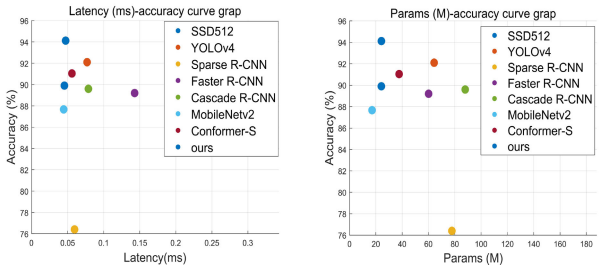
The proposed model consists of two parts, namely, the BAC module and the FSAR module. To analyze the importance of each part, an ablation experiment is performed in Table I.

TABLE III
COMPARISON RESULTS FOR DIFFERENT METHODS OF SA

method	mAP**(%)**	$Params$**(M)**	$Flops$**(G)**
SA	91.04	24.24	5.2
FSAR	92.56	24.24	5.2

TABLE IV
MAP (%) FOR EACH CATEGORY OF THE NWPU-10 DATASET

method	mAP**(%)**	FPS	$Params$**(M)**
YOLOv4	92.1	13.5	64.13
Spares R-CNN	76.4	16.8	77.8
Faster R-CNN	89.2	7.0	60.19
Cascade R-CNN	89.6	12.7	87.9
SSD512	89.90	23.3	24.2
MobileNetV2	87.67	22.5	17.13
MobileVit	79.67	**43.4**	**7.94**
Conformer-S	91.04	17.9	24.24
ShuffleNet	90.92	22.2	12.11
Ours	**94.12**	24.24	21.2



(a) (b)

Fig. 6. Comparison of mainstream detection frameworks on the NWPU-10 dataset. (a) Latency (ms)—accuracy (%). (b) Params (M)—accuracy (%).

1) BAC Module: Compared with the baseline, the BAC module improves the detection accuracy by 2.08% (from 91.04% to 93.12%). This result shows that the BAC module effectively extracts the contextual information of the object and improves the feature extraction capability of the model. Note that the BAC module achieves performance growth without adding parameters.

2) FSAR Module: This module improves the detection accuracy of the baseline model to 92.56%. This indicates that the proposed FSAR module effectively extracts global information. With this lightweight SA module, the feature information is enriched, which is beneficial to the feature extraction of the object.

To further validate the effectiveness of the inner design for each module, we conduct experiments for each module separately in this section.

To evaluate the effectiveness of the Sobel algorithm in the BAC module, we use FFT, Robert, and Canny operators instead of the Sobel algorithm to conduct comparison experiments, which are shown in Table II. The experimental results indicate that using the Sobel operator as the boundary convolution kernel achieves better detection accuracy without the need for additional parameters and computations.

Similar to the BAC module, we also make comparison experiments to compare the performance using different SA

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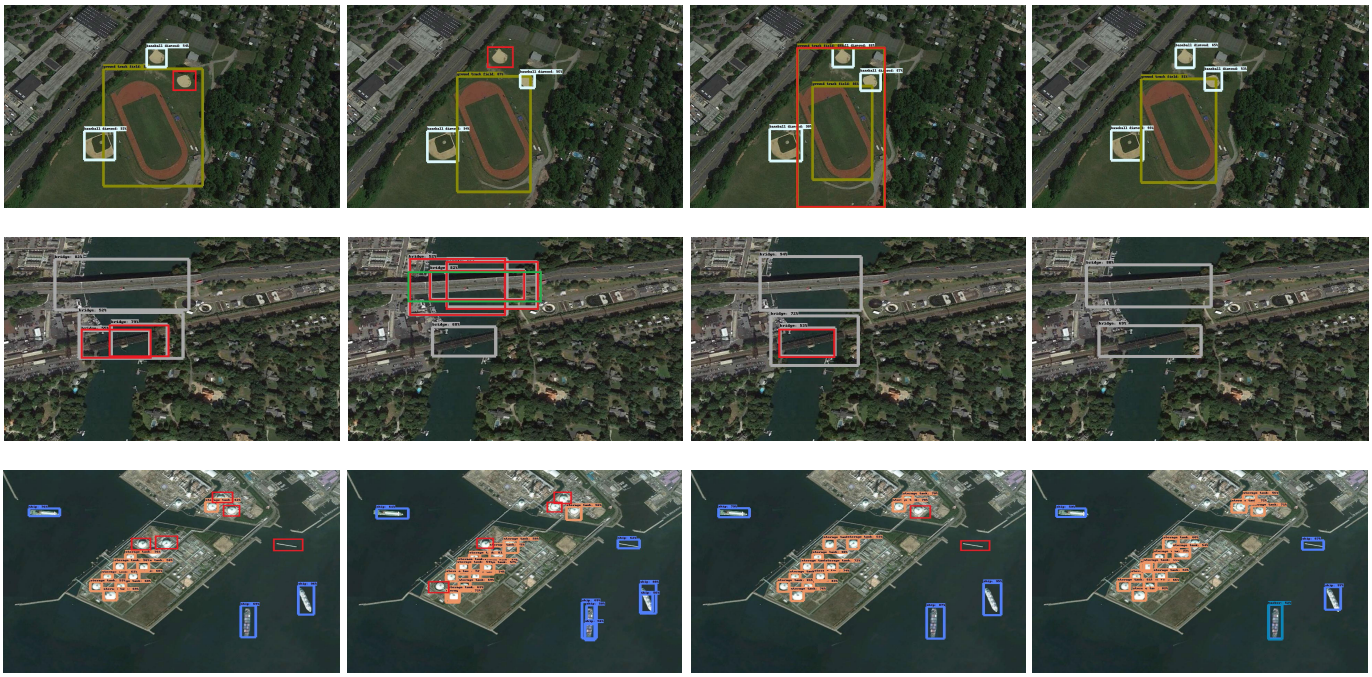


Fig. 7. Visualization of different models in the NWPU-10 dataset. Predicted bounding boxes with different colors correspond to different target categories, where red indicates a false detection and green indicates a correct detection frame. (a) YOLOv4. (b) MobileNetv2. (c) Conformer-S. (d) Ours.

TABLE V
MAP (%) FOR EACH CATEGORY OF THE RSOD DATASET

method	mAP**(%)**	FPS	$Params$**(M)**
YOLOv4	90.17	13.5	64.13
Spares R-CNN	83.00	16.8	77.8
Faster R-CNN	92.90	7.0	60.19
Cascade R-CNN	92.80	12.7	87.9
SSD512	92.30	23.3	24.2
MobileNetV2	93.65	22.5	17.13
MobileVit	90.24	**43.4**	**7.94**
Conformer-S	91.62	17.9	24.24
ShuffleNet	91.34	22.2	12.11
Ours	**95.58**	24.24	21.2

TABLE VI
MAP (%) FOR EACH CATEGORY OF THE LEVIR DATASET

method	mAP**(%)**	FPS	$Params$**(M)**
YOLOv4	81.56	13.5	64.13
Spares R-CNN	71.90	16.8	77.8
Faster R-CNN	77.50	7.0	60.19
Cascade R-CNN	78.70	12.7	87.9
SSD512	78.74	23.3	24.2
MobileNetV2	80.76	22.5	17.13
MobileVit	77.76	**43.4**	**7.94**
Conformer-S	83.19	17.9	24.24
ShuffleNet	82.60	22.2	12.11
Ours	**83.19**	24.24	21.2

mechanism, i.e., SA (traditional SA mechanism) and our FSAR, which is shown in Table III. The results show that our FSAR effectively extracts global information and improves the accuracy of the model.

D. Comparison With the State-of-the-Art Models

To evaluate the effectiveness and superiority of the proposed method, we conduct sufficient comparison experiments on the remote sensing benchmark datasets with many SOTA detectors, including mainstream object detection algorithms and lightweight detectors. In addition, we use mean average precision (mAP) and model parameters as evaluation metrics to comprehensively analyze the performance of the proposal algorithm.

1) Results on NWPU-10: The comparative results on the NWPU-10 dataset, as shown in Table IV and Fig. 6, demonstrate that our approach achieves the highest mAP compared with the suboptimal (mAP: 92.10%→94.12%). Despite not having the optimal latency, as observed in cases like SSD512 surpassing us, it is noteworthy that its accuracy is comparatively lower. While our model is not the most lightweight, with MobileVit surpassing us in terms of model size, its performance is relatively inferior. We have achieved a superior balance between model size and accuracy.

2) Results on RSOD: As shown in Table V, our method exhibits a significant advantage over mainstream detectors and lightweight detectors on the RSOD dataset. This includes both single-stage and two-stage methods, as well as hybrid-structured detectors. Achieving mAP of 95.58%, our method excels in striking a superior balance between model performance and size.

3) Results on LEVIR: As shown in Table VI, our approach has demonstrated outstanding performance in terms of accuracy, achieving an mAP of 84.13%. In comparison, the

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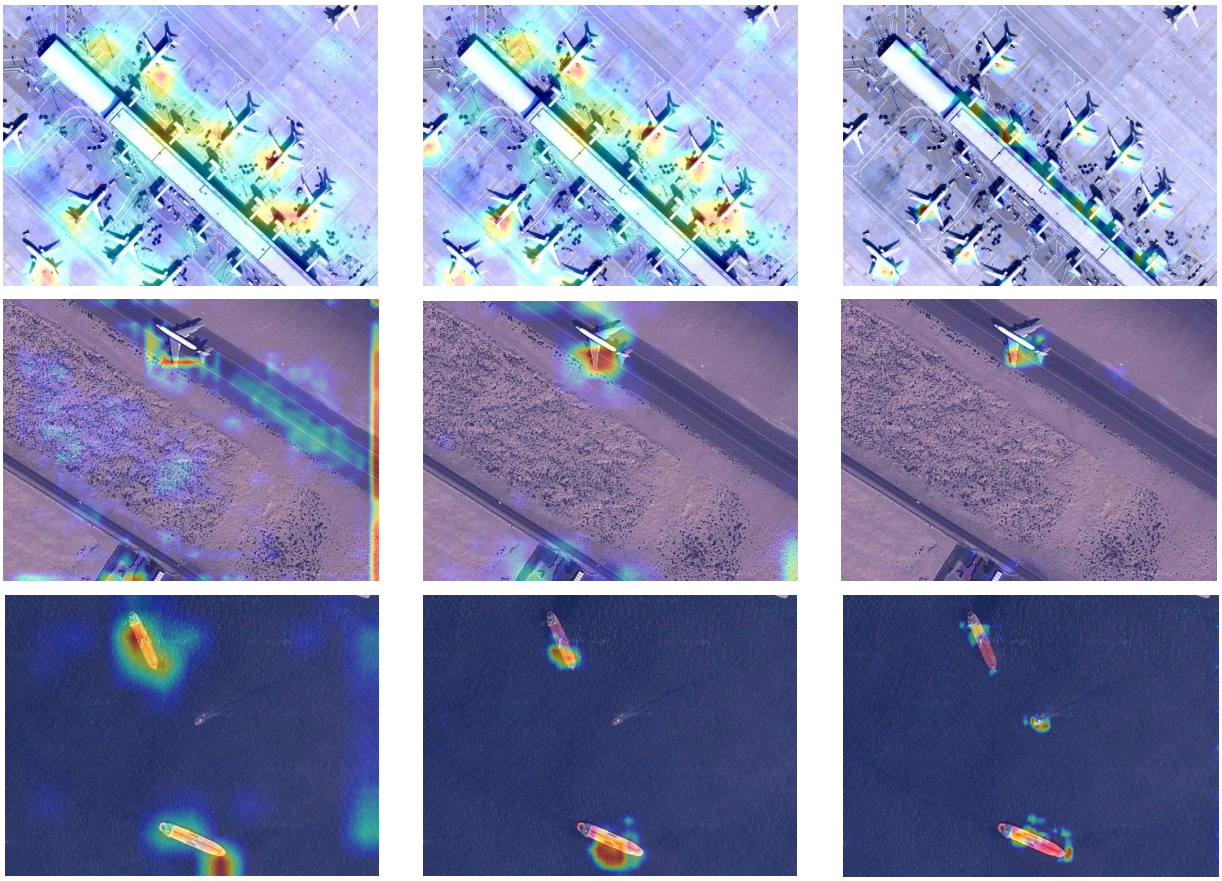


Fig. 8. Visual feature heat maps of the MobileNetv2, Conformer-S, and the proposed method on the LEVIR dataset. (a) MobileNetv2. (b) Conformer-S. (c) Ours.

TABLE VII
MAP (%) FOR EACH CATEGORY OF THE DOTAV1.0 DATASET

method	mAP**(%)**	FPS	$Params$**(M)**
YOLOv4	62.30	13.5	64.13
Faster R-CNN	60.64	7.0	60.19
Spares R-CNN	60.20	16.8	77.8
Cascade R-CNN	61.21	12.7	87.9
SSD512	62.4	23.3	24.2
MobileNetV2	56.91	22.5	17.13
MobileVit	59.8	**43.4**	**7.94**
Conformer-S	63.40	17.9	24.24
ShuffleNet	57.73	22.2	12.11
Ours	**66.00**	21.2	22.24

accuracy of the single-stage YOLO V4 and two-stage Faster R-CNN is only 81.56% and 77.50%, respectively. Even in comparison to lightweight detectors (MobileVit, SSD512), our method outperforms them significantly.

4) Results on DOTA: As indicated in Table VII, our method demonstrates outstanding accuracy across every category, achieving an mAP of 66.0%. In comparison, among mainstream detectors, the mAP of the single-stage YOLOv4 and two-stage Faster R-CNN are only 62.00% and 62.31%, respectively. In lightweight detection, model performance is relatively poor, with MobileNetV2 achieving only 56.91% mAP. These models have not been able to effectively balance performance and accuracy. This observation underscores the advantages of our approach.

In summary, compared with mainstream detection frameworks and lightweight models, our approach exhibits excellent performance across various datasets. It is noteworthy that our method achieves optimal accuracy results with a smaller parameter size of 22.4 M, further validating the robustness, efficiency, and lightweight nature of our approach in achieving a superior balance between model size and precision.

E. Visualization Analysis

1) Qualitative Analysis From Detection Results: On the NWPU-10 dataset, we qualitatively compare the detection results of YOLOv4, MobileNetv2, Conformer-S, and the proposed method. As shown in Fig. 7, our method successfully obtains the correct detection frames than other compared methods when facing multiscale images. Second, due to multiple bounding boxes of the same object are generated for these compared methods, leading to repeated detection problems, where the MobileNet V2 framework even leads to missed detection. Fortunately, our method successfully avoids these problems. Finally, when the image contains many targets and

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have complex background, other methods cause missed detection phenomenon, due to the absence of effective information, thus ignoring some targets in the complex background. In contrast, our method effectively solves these problems. The BAC module can effectively deal with multiscale problems by extracting contextual information. Furthermore, the FSAR module makes a perfect prevention for complex contextual problems by extracting effective information, and therefore, our method shows better results.

2) Qualitative Analysis From Feature Heat Maps: As shown in Fig. 8, we use a heatmap of detections from the same framework to highlight the differences between our approach and others, emphasizing our strengths. This visualization offers a clear and intuitive view of the results. It is evident that MobileNet V2 and Conformers-S cannot accurately highlight multiple targets in a complex background, resulting in terrible detection results. In contrast, our method, enhanced by the BAC and FSAR modules, effectively extracts valuable feature information, especially for fine-target details.

V. Conclusion

In this article, we propose an effective and lightweight detection framework of hybrid network for remote sensing images. Global and local features of the object are extracted by CNN and transformer, respectively, to obtain a more efficient and richer feature representation. A BAC module is introduced to improve the detection accuracy by extracting the contextual information of the object. In addition, we propose an FSAR module to improve the extraction of feature information at different low frequencies. The experimental results show that our model achieves impressive performance with fewer parameters compared with SOTA detectors.

In future work, we will focus more on the application of combined CNN and transformer networks in computer vision, especially in object detection. Other neural network compression techniques such as knowledge distillation and structural reparameterization will also be explored.

References

[1] R. Girshick, J. Donahue, T. Darrell, and J. Malik, "Rich feature hierarchies for accurate object detection and semantic segmentation," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2014, pp. 580–587.

[2] R. Girshick, "Fast R-CNN," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Dec. 2015, pp. 1440–1448.

[3] S. Ren, K. He, R. Girshick, and J. Sun, "Faster R-CNN: Towards real-time object detection with region proposal networks," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 28, 2015, pp. 91–99.

[4] J. Pang, K. Chen, J. Shi, H. Feng, W. Ouyang, and D. Lin, "Libra R-CNN: Towards balanced learning for object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 821–830.

[5] Z. Cai and N. Vasconcelos, "Cascade R-CNN: Delving into high quality object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 6154–6162.

[6] J. Redmon and A. Farhadi, "YOLOv3: An incremental improvement," 2018, *arXiv:1804.02767*.

[7] A. Bochkovskiy, C.-Y. Wang, and H.-Y. Mark Liao, "YOLOv4: Optimal speed and accuracy of object detection," 2020, *arXiv:2004.10934*.

[8] Q. Chen, Y. Wang, T. Yang, X. Zhang, J. Cheng, and J. Sun, "You only look one-level feature," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 13034–13043.

[9] Z. Ge, S. Liu, F. Wang, Z. Li, and J. Sun, "YOLOX: Exceeding YOLO series in 2021," 2021, *arXiv:2107.08430*.

[10] C.-Y. Wang, A. Bochkovskiy, and H.-Y. M. Liao, "YOLOv7: Trainable bag-of-freebies sets new state-of-the-art for real-time object detectors," 2022, *arXiv:2207.02696*.

[11] C. Li et al., "YOLOv6: A single-stage object detection framework for industrial applications," 2022, *arXiv:2209.02976*.

[12] K. Duan, S. Bai, L. Xie, H. Qi, Q. Huang, and Q. Tian, "CenterNet: Keypoint triplets for object detection," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 6568–6577.

[13] A. Dosovitskiy et al., "An image is worth 16×16 words: Transformers for image recognition at scale," 2020, *arXiv:2010.11929*.

[14] Z. Liu et al., "Swin transformer: Hierarchical vision transformer using shifted windows," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 9992–10002.

[15] Z. Liu et al., "Swin transformer v2: Scaling up capacity and resolution," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 11999–12009.

[16] Q. Zhang and Y.-B. Yang, "ResT: An efficient transformer for visual recognition," in *Proc. Adv. Neural Inform. Process. Syst.*, vol. 34, 2021, pp. 15475–15485.

[17] L. Yuan et al., "Tokens-to-token ViT: Training vision transformers from scratch on ImageNet," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 538–547.

[18] Z. Peng et al., "Conformer: Local features coupling global representations for visual recognition," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 357–366.

[19] B. Wu, A. Wan, F. Iandola, P. H. Jin, and K. Keutzer, "SqueezeDet: Unified, small, low power fully convolutional neural networks for real-time object detection for autonomous driving," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jul. 2017, pp. 446–454.

[20] Y. Li, J. Li, W. Lin, and J. Li, "Tiny-DSOD: Lightweight object detection for resource-restricted usages," 2018, *arXiv:1807.11013*.

[21] R. J. Wang, X. Li, and C. X. Ling, "Pelee: A real-time object detection system on mobile devices," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 31, 2018, pp. 1–10.

[22] X. Zhu, W. Su, L. Lu, B. Li, X. Wang, and J. Dai, "Deformable DETR: Deformable transformers for end-to-end object detection," 2020, *arXiv:2010.04159*.

[23] S. Zheng et al., "Rethinking semantic segmentation from a sequence-to-sequence perspective with transformers," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 6877–6886.

[24] G. Cheng, P. Zhou, and J. Han, "Learning rotation-invariant convolutional neural networks for object detection in VHR optical remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 12, pp. 7405–7415, Dec. 2016.

[25] Z. Zou and Z. Shi, "Random access memories: A new paradigm for target detection in high resolution aerial remote sensing images," *IEEE Trans. Image Process.*, vol. 27, no. 3, pp. 1100–1111, Mar. 2018.

[26] Y. Long, Y. Gong, Z. Xiao, and Q. Liu, "Accurate object localization in remote sensing images based on convolutional neural networks," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 5, pp. 2486–2498, May 2017.

[27] G.-S. Xia et al., "DOTA: A large-scale dataset for object detection in aerial images," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 3974–3983.

[28] G. Gao, "An improved scheme for target discrimination in high-resolution SAR images," *IEEE Trans. Geosci. Remote Sens.*, vol. 49, no. 1, pp. 277–294, Jan. 2011.

[29] H. Guo, X. Yang, N. Wang, B. Song, and X. Gao, "A rotational libra R-CNN method for ship detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 8, pp. 5772–5781, Aug. 2020.

[30] M. Kang, X. Leng, Z. Lin, and K. Ji, "A modified faster R-CNN based on CFAR algorithm for SAR ship detection," in *Proc. Int. Workshop Remote Sens. Intell. Process. (RSIP)*, May 2017, pp. 1–4.

[31] X. Zhou et al., "Edge-guided recurrent positioning network for salient object detection in optical remote sensing images," *IEEE Trans. Cybern.*, vol. 53, no. 1, pp. 539–552, Jan. 2023.

[32] E. Liu, Y. Zheng, B. Pan, X. Xu, and Z. Shi, "DCL-Net: Augmenting the capability of classification and localization for remote sensing object detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 9, pp. 7933–7944, Sep. 2021.

[33] Z. Yuan, Z. Liu, C. Zhu, J. Qi, and D. Zhao, "Object detection in remote sensing images via multi-feature pyramid network with receptive field block," *Remote Sens.*, vol. 13, no. 5, p. 862, Feb. 2021.

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[34] G. Shi, J. Zhang, J. Liu, C. Zhang, C. Zhou, and S. Yang, "Global context-augmented objection detection in VHR optical remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 12, pp. 10604–10617, Dec. 2021.

[35] J. Hu, L. Shen, and G. Sun, "Squeeze-and-excitation networks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 7132–7141.

[36] Y. Li, Q. Huang, X. Pei, L. Jiao, and R. Shang, "RADet: Refine feature pyramid network and multi-layer attention network for arbitrary-oriented object detection of remote sensing images," *Remote Sens.*, vol. 12, no. 3, p. 389, Jan. 2020.

[37] Q. Li, Y. Chen, and Y. Zeng, "Transformer with transfer CNN for remote-sensing-image object detection," *Remote Sens.*, vol. 14, no. 4, p. 984, Feb. 2022.

[38] J. Chen, H. Hong, B. Song, J. Guo, C. Chen, and J. Xu, "MDCT: Multi-kernel dilated convolution and transformer for one-stage object detection of remote sensing images," *Remote Sens.*, vol. 15, no. 2, p. 371, Jan. 2023.

[39] G. Xu, T. Song, X. Sun, and C. Gao, "TransMIN: Transformer-guided multi-interaction network for remote sensing object detection," *IEEE Geosci. Remote Sens. Lett.*, vol. 20, pp. 1–5, 2023.

[40] Y. Zheng, P. Sun, Z. Zhou, W. Xu, and Q. Ren, "ADT-Det: Adaptive dynamic refined single-stage transformer detector for arbitrary-oriented object detection in satellite optical imagery," *Remote Sens.*, vol. 13, no. 13, p. 2623, Jul. 2021.

[41] G. Li, Z. Liu, D. Zeng, W. Lin, and H. Ling, "Adjacent context coordination network for salient object detection in optical remote sensing images," *IEEE Trans. Cybern.*, vol. 53, no. 1, pp. 526–538, Jan. 2023.

[42] A. G. Howard et al., "MobileNets: Efficient convolutional neural networks for mobile vision applications," 2017, *arXiv:1704.04861*.

[43] M. Sandler, A. Howard, M. Zhu, A. Zhmoginov, and L.-C. Chen, "MobileNetV2: Inverted residuals and linear bottlenecks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 4510–4520.

[44] A. Howard et al., "Searching for MobileNetV3," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 1314–1324.

[45] X. Zhang, X. Zhou, M. Lin, and J. Sun, "ShuffleNet: An extremely efficient convolutional neural network for mobile devices," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 6848–6856.

[46] N. Ma, X. Zhang, H.-T. Zheng, and J. Sun, "ShuffleNet v2: Practical guidelines for efficient CNN architecture design," in *Proc. Eur. Conf. Comput. Vis.*, 2018, pp. 116–131.

[47] K. Han, Y. Wang, Q. Tian, J. Guo, C. Xu, and C. Xu, "GhostNet: More features from cheap operations," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 1577–1586.

[48] Y. Li, S. Zhang, and W.-Q. Wang, "A lightweight faster R-CNN for ship detection in SAR images," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[49] L.-Q. Zhou and J.-C. Piao, "A lightweight YOLOv4 based SAR image ship detection," in *Proc. IEEE 4th Int. Conf. Comput. Commun. Eng. Technol. (CCET)*, Aug. 2021, pp. 28–31.

[50] Y. Jiang, Z. Tan, J. Wang, X. Sun, M. Lin, and H. G. Li, "A heavy-neck paradigm for object detection," in *Proc. Int. Conf. Learn. Represent.*, vol. 4, 2021, pp. 1–17.

[51] R. Child, S. Gray, A. Radford, and I. Sutskever, "Generating long sequences with sparse transformers," 2019, *arXiv:1904.10509*.

[52] I. Beltagy, M. E. Peters, and A. Cohan, "Longformer: The long-document transformer," 2020, *arXiv:2004.05150*.

[53] G. Zhang, Z. Luo, Z. Tian, J. Zhang, X. Zhang, and S. Lu, "Towards efficient use of multi-scale features in transformer-based object detectors," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2023, pp. 6206–6216.

[54] J. Guo et al., "CMT: Convolutional neural networks meet vision transformers," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 12165–12175.

[55] K. Li et al., "UniFormer: Unifying convolution and self-attention for visual recognition," 2022, *arXiv:2201.09450*.

[56] Z. Peng et al., "Conformer: Local features coupling global representations for recognition and detection," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 8, pp. 9454–9468, Aug. 2023, doi: 10.1109/TPAMI.2023.3243048.

[57] S. Mehta and M. Rastegari, "MobileViT: Light-weight, general-purpose, and mobile-friendly vision transformer," 2021, *arXiv:2110.02178*.

[58] Y. Yang et al., "Adaptive knowledge distillation for lightweight remote sensing object detectors optimizing," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–15, 2022, Art. no. 5623715, doi: 10.1109/TGRS.2022.3175213.



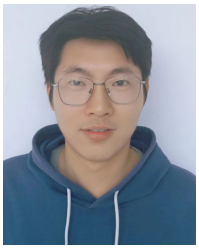
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